Software Requirements Specification

for

FitTrack Pro

Fitness Tracking System

Version 1.0 – Approved

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Revision History

Name	Date	Reason For Changes	Version
FitTrack Dev Team	May 2025	Initial Release	1.0

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1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) document describes the functional and non-functional requirements for FitTrack Pro, a comprehensive fitness tracking system. The document covers Version 1.0 of the application, targeting both mobile and web platforms.

This SRS is intended to guide developers, testers, designers, and stakeholders throughout the software development lifecycle. Specifically, the document defines:

- The goals and objectives of the FitTrack Pro system
- All functional requirements (features and behaviors)
- Non-functional requirements (performance, security, usability)
- External interface requirements (user, hardware, software, and communications)
- Constraints and assumptions affecting development

1.2 Document Conventions

Convention	Meaning
Shall	Mandatory requirement – must be implemented
Should	Recommended – highly desired but not strictly mandatory
May	Optional – can be implemented if resources allow
FR-XX	Functional Requirement identifier
NFR-XX	Non-Functional Requirement identifier
Bold text	Section headings and key terms
High / Medium / Low	Feature priority levels

1.3 Project Scope

FitTrack Pro is a cross-platform fitness tracking application designed to help users monitor their physical activity, nutrition, sleep, and overall health metrics. The system is intended to:

- Track daily workouts, steps, calories burned, and heart rate
- Log nutritional intake and provide macro/micro-nutrient breakdowns
- Monitor sleep patterns and provide quality scores
- Enable goal setting and milestone tracking
- Integrate with popular wearable devices (smartwatches, fitness bands)
- Support social sharing and community challenges

1.4 References

- IEEE Std 830-1998 – Recommended Practice for Software Requirements Specifications

- HL7 FHIR Standards for Health Data Interoperability
- Apple HealthKit Developer Documentation
- Google Fit REST API Documentation
- GDPR Compliance Guidelines (2016/679)
- HIPAA Security Rules for Health Applications

2. Overall Description

2.1 Product Perspective

FitTrack Pro is a standalone mobile and web application that serves as a centralized personal fitness hub. The system follows a client-server architecture:

- Mobile Clients (iOS & Android) – primary user-facing interface
- Web Application – browser-based access for detailed analytics and account management
- Backend API Server – processes business logic and data storage
- Machine Learning Engine – generates personalized recommendations
- Third-Party Integrations – wearables, nutrition databases, and health platforms

2.2 Product Features

#	Feature	Description
F1	Activity Tracking	Record workouts, steps, distance, and calories burned in real-time
F2	Nutrition Logging	Log meals and track macro/micro-nutrient intake with barcode scanning
F3	Sleep Monitoring	Analyze sleep duration, stages, and quality score automatically
F4	Goal Management	Set, track, and achieve personalized fitness and wellness goals
F5	Dashboard & Analytics	Visualize trends with charts, heatmaps, and weekly/monthly reports
F6	AI Recommendations	Receive personalized workout plans and nutrition tips based on user data
F7	Wearable Integration	Sync data from Apple Watch, Fitbit, Garmin, Samsung Galaxy Watch
F8	Social & Community	Join challenges, share achievements, and follow friends
F9	Notifications & Reminders	Smart reminders for workouts, hydration, meals, and rest
F10	User Profile & Settings	Manage personal information, health metrics, and app preferences

2.3 User Classes and Characteristics

User Class	Technical Level	Frequency of Use	Key Permissions
Regular User	Beginner to Intermediate	Daily	Log activities, view own data, set goals
Premium User	Intermediate	Daily	All Regular features + AI plans, advanced analytics
Personal Trainer	Intermediate to Advanced	Daily	Manage client profiles, assign plans, track progress

Administrator	Advanced / Technical	Weekly	Full system access, user management, reports
Guest / Visitor	Any	One-time / Occasional	Read-only landing page and feature preview

2.4 Operating Environment

Mobile Application

- iOS 15.0 or later (iPhone and iPad)
- Android 10.0 (API level 29) or later
- Minimum 2GB RAM, 100MB free storage

Web Application

- Supported browsers: Google Chrome 100+, Mozilla Firefox 100+, Safari 15+, Microsoft Edge 100+
- Minimum screen resolution: 1280 x 720

Backend Infrastructure

- Cloud: Amazon Web Services (AWS)
- Database: PostgreSQL 14+ and Redis 7+ for caching
- Server: Node.js 18+ runtime, OS: Ubuntu 22.04 LTS

2.5 Design and Implementation Constraints

- The mobile application shall be built using React Native for cross-platform compatibility
- The web frontend shall use React.js with TypeScript
- The backend API shall be built using Node.js with Express.js
- All data at rest shall be encrypted using AES-256
- All data in transit shall use HTTPS with TLS 1.2 or higher
- The system shall comply with GDPR for users in the European Union
- The AI recommendation engine shall be developed using Python with TensorFlow

3. System Features

3.1 Activity Tracking

3.1.1 Description and Priority

The Activity Tracking feature allows users to record and monitor all forms of physical activity, including workouts, daily steps, distance traveled, and calories burned. Priority: High

3.1.2 Stimulus / Response Sequences

Stimulus	System Response
User starts a workout session	System begins tracking in real-time (GPS, heart rate, timer)
User pauses workout	System pauses all sensors and saves intermediate data
User completes workout	System saves session, calculates summary, and updates dashboard
User logs a manual activity	System validates input, saves entry, and recalculates daily totals
Wearable syncs step data	System merges wearable data and updates step count

3.1.3 Functional Requirements

- FR-1.1: The system shall allow users to start, pause, resume, and stop workout sessions.
- FR-1.2: The system shall track GPS route, distance, pace, and duration for outdoor activities.
- FR-1.3: The system shall support over 50 predefined workout types.
- FR-1.4: The system shall calculate calories burned using the user's weight, age, gender, and activity intensity.
- FR-1.5: The system shall sync step count and heart rate data from connected wearable devices.
- FR-1.6: The system shall allow manual entry of past activities with date and duration.
- FR-1.7: The system shall display a post-workout summary with key metrics.
- FR-1.8: The system shall store a complete history of all workout sessions.

3.2 Nutrition Logging

3.2.1 Description and Priority

Enables users to record daily food and drink consumption, view detailed nutritional breakdowns, and track calorie intake against their goals. Priority: High

3.2.2 Functional Requirements

- FR-2.1: The system shall allow users to log meals by searching a database of 1,000,000+ food items.
- FR-2.2: The system shall support barcode scanning to auto-populate nutritional information.
- FR-2.3: The system shall display daily totals for calories, protein, carbohydrates, fat, fiber, and micronutrients.
- FR-2.4: The system shall allow users to create custom foods and save personal recipes.

- FR-2.5: The system shall support meal planning by scheduling meals in advance.
- FR-2.6: The system shall provide a water intake tracker with daily goal and reminders.
- FR-2.7: The system shall calculate net calories (calories consumed minus calories burned).

3.3 Sleep Monitoring

Priority: Medium

- FR-3.1: The system shall automatically detect sleep start and end times via wearable sensors.
- FR-3.2: The system shall categorize sleep into stages: Awake, Light, Deep, and REM.
- FR-3.3: The system shall calculate a nightly Sleep Quality Score (0–100).
- FR-3.4: The system shall allow manual logging of sleep hours if no wearable is connected.
- FR-3.5: The system shall display weekly and monthly sleep trend charts.
- FR-3.6: The system shall send a bedtime reminder notification based on the user's sleep goal.

3.4 Goal Management

Priority: High

- FR-4.1: The system shall allow users to set goals for weight, daily steps, weekly workouts, calorie intake, and sleep duration.
- FR-4.2: The system shall display real-time progress toward each active goal.
- FR-4.3: The system shall notify users when they achieve a goal or milestone.
- FR-4.4: The system shall allow users to edit or delete existing goals.
- FR-4.5: The system shall provide goal completion history and streaks.
- FR-4.6: The system shall suggest next goals based on user progress and AI analysis.

3.5 Dashboard and Analytics

Priority: High

- FR-5.1: The system shall display a daily summary card with steps, calories, active minutes, and sleep score.
- FR-5.2: The system shall provide weekly and monthly trend charts for all major metrics.
- FR-5.3: The system shall support customizable dashboard widgets.
- FR-5.4: The system shall generate downloadable PDF health reports.
- FR-5.5: The system shall display an activity heatmap showing consistency over time.

3.6 AI-Powered Recommendations

Priority: Medium

- FR-6.1: The system shall generate a personalized weekly workout plan based on fitness level, goals, and available equipment.
- FR-6.2: The system shall provide nutrition recommendations aligned with the user's dietary goals.
- FR-6.3: The system shall suggest rest or recovery days when overtraining patterns are detected.

- FR-6.4: The system shall update recommendations weekly based on performance and feedback.
- FR-6.5: The system shall allow users to rate recommendations to improve accuracy.

3.7 Wearable Device Integration

Priority: High

- FR-7.1: The system shall support integration with Apple Watch via HealthKit.
- FR-7.2: The system shall support integration with Fitbit via Fitbit Web API.
- FR-7.3: The system shall support integration with Garmin devices via Garmin Connect API.
- FR-7.4: The system shall support integration with Samsung Galaxy Watch via Samsung Health SDK.
- FR-7.5: The system shall sync heart rate, steps, sleep, and workout data automatically every 15 minutes.
- FR-7.6: The system shall notify users if a sync fails and provide retry options.

3.8 Social Features and Community Challenges

Priority: Medium

- FR-8.1: The system shall allow users to follow other users and view their public activity feed.
- FR-8.2: The system shall allow users to participate in timed community challenges.
- FR-8.3: The system shall display a leaderboard for active challenges.
- FR-8.4: The system shall allow users to share achievements to social media platforms.
- FR-8.5: The system shall allow users to set their profile to Public, Friends Only, or Private.

3.9 Notifications and Reminders

Priority: Medium

- FR-9.1: The system shall send daily step goal reminders based on user progress.
- FR-9.2: The system shall send workout schedule reminders at user-defined times.
- FR-9.3: The system shall send hydration reminders at configurable intervals.
- FR-9.4: The system shall send achievement notifications when milestones are reached.
- FR-9.5: The system shall allow users to fully customize notification preferences.
- FR-9.6: The system shall respect device Do Not Disturb settings.

4. External Interface Requirements

4.1 User Interfaces

FitTrack Pro shall provide an intuitive, modern GUI consistent across mobile and web platforms. The interface shall include:

- Onboarding Flow – account creation, profile setup, goal configuration wizard
- Login / Sign-Up Screen – email/password and OAuth (Google, Apple)
- Main Dashboard – summary cards, progress rings, and quick-add shortcuts
- Activity Screen – workout start, live tracking view, and history log
- Nutrition Screen – meal diary, food search, barcode scanner, and macro ring
- Sleep Screen – nightly summary card, quality score, and trend chart
- Goals Screen – active goals, progress bars, and milestone badges
- Analytics Screen – interactive charts, heatmaps, and report download

4.2 Hardware Interfaces

Hardware	Purpose	Interface Method
Smartwatch / Fitness Band	Heart rate, steps, sleep data	Bluetooth LE / Vendor API
Smartphone Camera	Food barcode scanning	Native Camera API
GPS Chip (Mobile)	Outdoor route tracking	CoreLocation / Location API

4.3 Software Interfaces

System / API	Version	Protocol	Purpose
Apple HealthKit	iOS 15+	Native SDK	Health data read/write on iOS
Google Fit API	v1	REST / OAuth 2.0	Health data sync on Android
Fitbit Web API	v1.2	REST / OAuth 2.0	Fitbit device data sync
Garmin Connect API	v1	REST / OAuth 1.0a	Garmin device data sync
USDA FoodData API	v1	REST / API Key	Nutritional data lookup
Firebase (FCM)	v9	HTTPS / FCM Protocol	Push notifications delivery
PostgreSQL	v14	TCP/IP	Primary data storage
Redis	v7	TCP/IP	Session caching and rate limiting
AWS S3	Current	HTTPS / SDK	Media and report file storage
AWS Cognito	Current	HTTPS / OAuth 2.0	Authentication and user management

4.4 Communications Interfaces

- HTTPS with TLS 1.2 or higher for all client-server communication
- WebSocket for real-time workout tracking data streams
- Bluetooth Low Energy (BLE) for direct wearable device communication
- RESTful API architecture with JSON data format for all API endpoints
- JWT (JSON Web Tokens) for authentication and session management
- OAuth 2.0 used for third-party service authentication
- API rate limiting: 1000 requests per hour per user

5. Other Nonfunctional Requirements

5.1 Performance Requirements

ID	Requirement	Target
NFR-1	API response time for dashboard load	< 2 seconds under normal load
NFR-2	Food search query response time	< 1 second for database queries
NFR-3	Real-time workout data update interval	Every 1 second during active session
NFR-4	Wearable data sync latency	< 15 minutes for automatic sync
NFR-5	Concurrent user support	Minimum 100,000 concurrent users
NFR-6	System uptime / availability	99.9% uptime (SLA)
NFR-7	Mobile app launch time	< 3 seconds cold start
NFR-8	Report PDF generation time	< 10 seconds for monthly report

5.2 Safety Requirements

- The system shall display a disclaimer that FitTrack Pro is not a substitute for professional medical advice.
- The system shall warn users if their logged calorie intake falls below 1,200 kcal/day (women) or 1,500 kcal/day (men).
- The system shall alert users if heart rate data from a wearable exceeds 190 bpm during a workout.
- The system shall recommend consulting a doctor before starting a new fitness plan if the user's BMI is above 35 or below 18.
- The system shall never recommend extreme fasting or crash diets based on user goals.

5.3 Security Requirements

- NFR-SEC-1: All user passwords shall be hashed using bcrypt with a minimum cost factor of 12.
- NFR-SEC-2: The system shall enforce multi-factor authentication (MFA) for all accounts.
- NFR-SEC-3: All health data shall be encrypted at rest using AES-256 encryption.
- NFR-SEC-4: The system shall implement GDPR-compliant data handling, including the right to deletion.
- NFR-SEC-5: The system shall log all authentication events and API access for audit purposes.
- NFR-SEC-6: The system shall automatically log out inactive sessions after 30 minutes.
- NFR-SEC-7: The system shall perform automated vulnerability scanning monthly.
- NFR-SEC-8: Personal data shall never be shared with third parties without explicit user consent.

5.4 Software Quality Attributes

Attribute	Requirement
Usability	New users shall be able to complete onboarding and log first activity within 5 minutes
Reliability	System shall recover from crashes automatically and restore user session without data loss
Maintainability	Codebase shall maintain 80%+ unit test coverage; modules shall be independently deployable
Portability	Application shall run on iOS 15+, Android 10+, and all major web browsers without code changes
Scalability	System architecture shall support horizontal scaling to handle 10x user growth without redesign
Testability	All API endpoints shall be documented and testable via Swagger / OpenAPI specification
Interoperability	System shall integrate with at least 4 major wearable platforms via their official APIs

6. UML Diagrams

This section contains the Unified Modeling Language (UML) diagrams derived from the requirements defined in this SRS. Five diagram types are included: Use Case, Activity, Sequence, State Machine, and Class Diagram.

6.1 Use Case Diagram

The use case diagram below illustrates all system actors and their interactions with the FitTrack Pro system. Actors include Regular User, Premium User, Personal Trainer, Administrator, Wearable Device, and Nutrition API. Use cases cover all ten core features from Section 3.

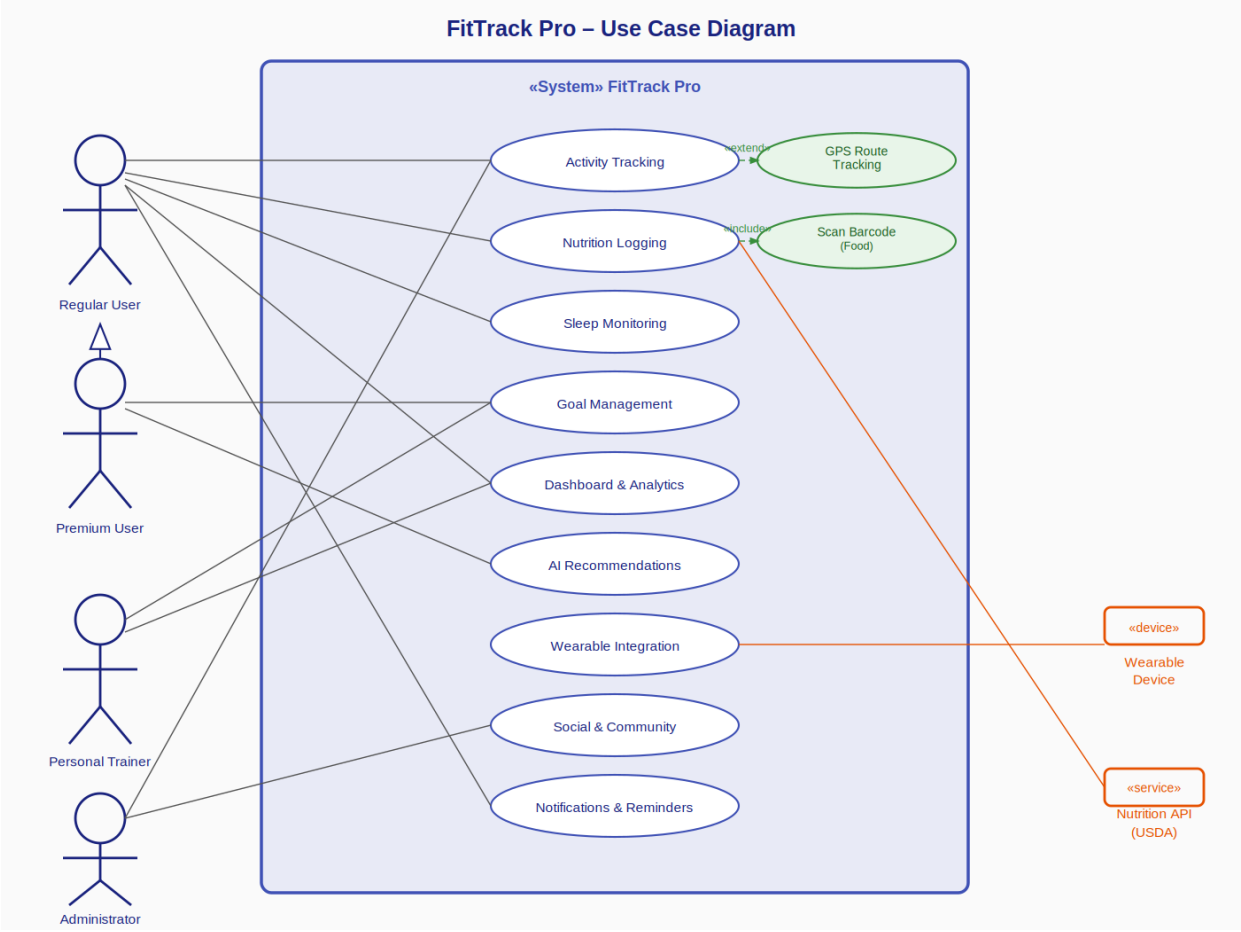


Figure 6.1 – FitTrack Pro Use Case Diagram

6.2 Activity Diagram – Workout Session Flow

This activity diagram models the complete workout session lifecycle: from opening the app and authentication, through selecting a workout type, starting parallel activities (GPS tracking and heart rate monitoring), pausing/resuming, stopping, and finally generating and displaying the post-workout summary. It directly reflects requirements FR-1.1 through FR-1.7.

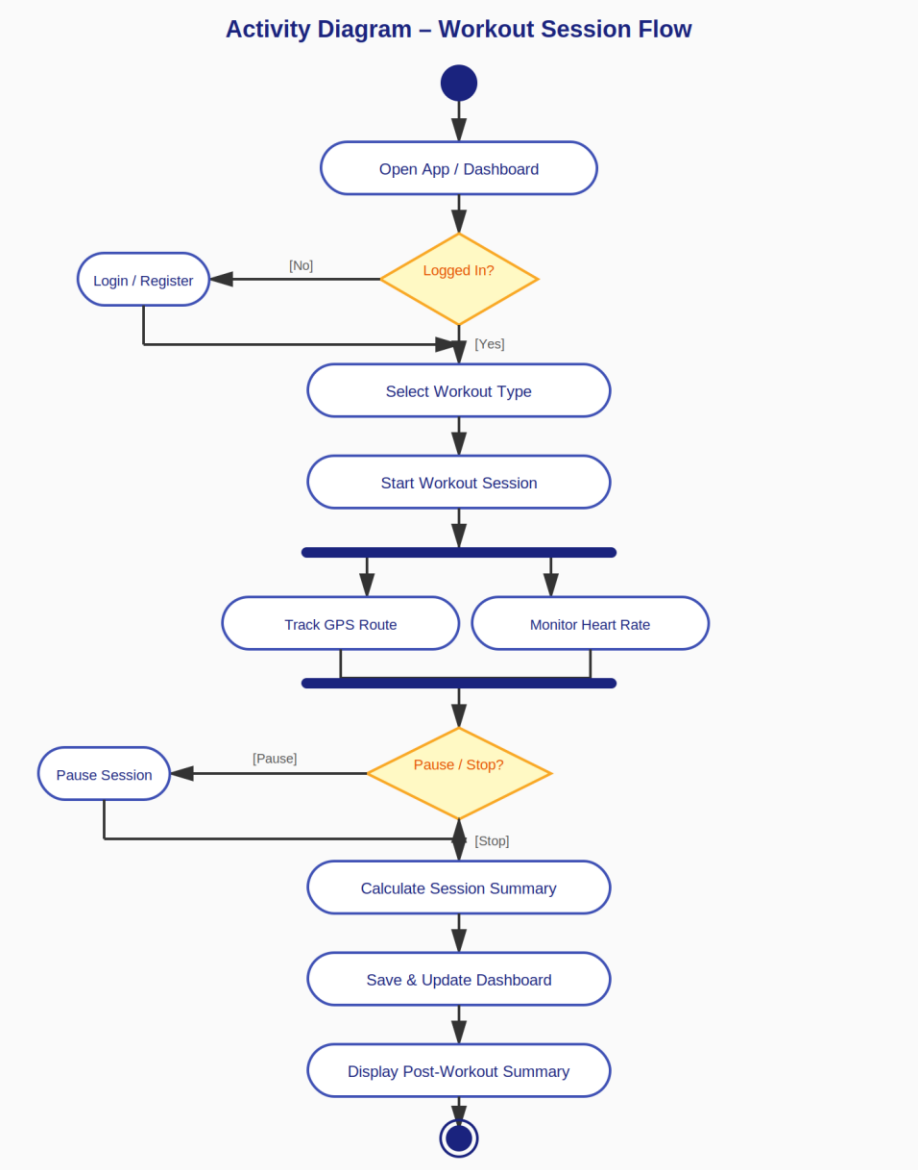


Figure 6.2 – Activity Diagram: Workout Session Flow

6.3 Sequence Diagram – User Login & Wearable Sync

This sequence diagram shows the time-ordered interactions between the User, Mobile App, Backend API, AWS Cognito (authentication), and the Wearable API. It covers login with JWT token issuance, wearable data sync (FR-7.5), and the background 15-minute sync loop with push notifications (FR-9.4).

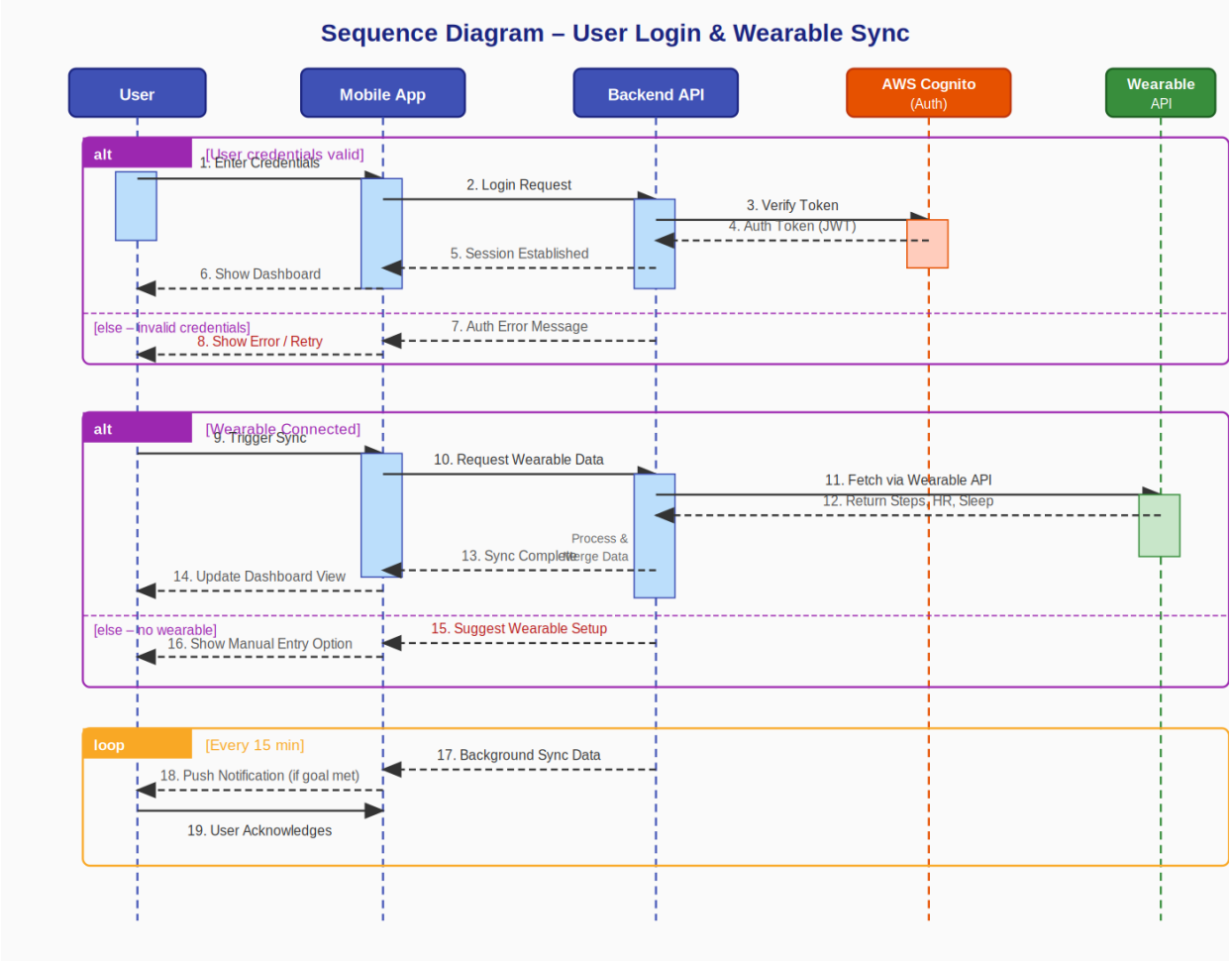


Figure 6.3 – Sequence Diagram: Login & Wearable Sync

6.4 State Machine Diagram – Workout Session States

This state machine diagram depicts all possible states of a WorkoutSession entity: IDLE, READY, ACTIVE, PAUSED, STOPPED, and SUMMARY. Transitions are triggered by user actions (startSession, pauseSession, stopSession) and system events (heart rate alert). This models the behavior described in FR-1.1 and FR-1.7.

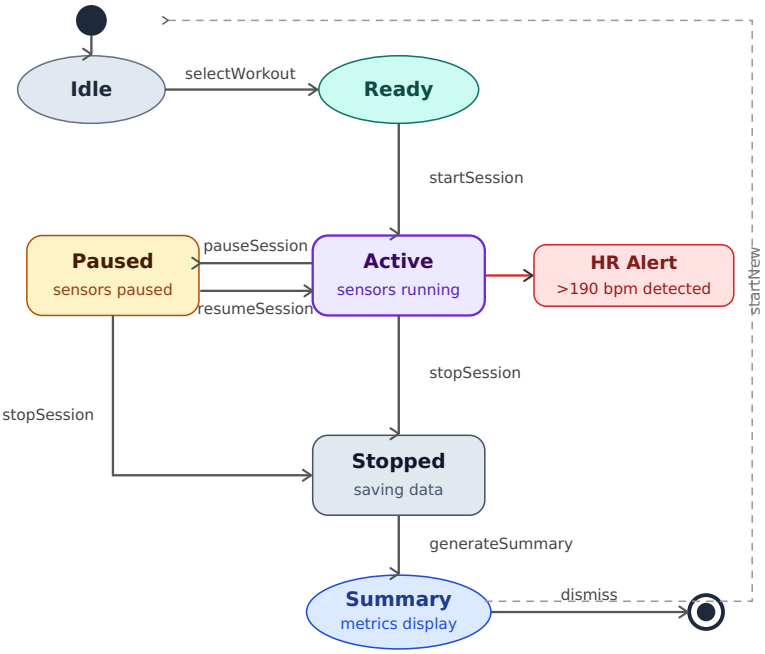


Figure 6.4 – State Machine Diagram: Workout Session

6.5 Class Diagram – System Data Model

The class diagram defines the main entities of the FitTrack Pro system and their relationships. The central User class is associated (1-to-many) with WorkoutSession, NutritionLog, SleepRecord, Goal, and WearableDevice. Each class exposes its key attributes and methods. This structural model supports all functional requirements in Section 3.

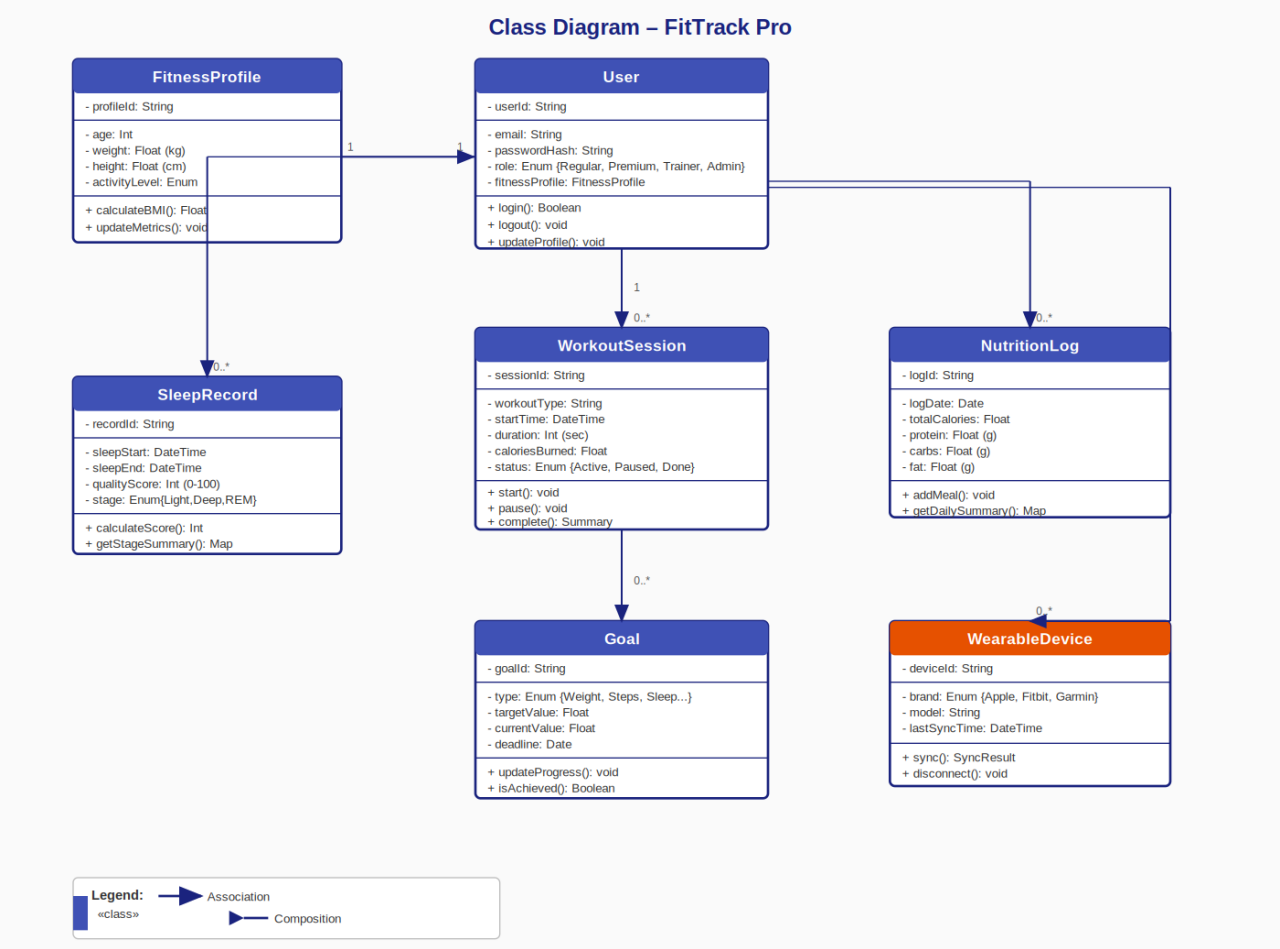


Figure 6.5 – Class Diagram: FitTrack Pro Data Model